

Problem Statement - Intelligent Complaint Classification

Consumer-support complaints can be slow and inconsistent to route manually. This project classifies free-text complaints into ten operational categories: account access, billing, customer service, delivery/shipping, fraud/unauthorized activity, general inquiry, product defect, refund/return, subscription cancellation, and warranty/repair.

The intended use is decision support for first-pass routing, not automated resolution. Support staff should review predictions, especially for fraud, account access, and refund cases.

The project objective is macro F1, so every category contributes equally. The final approach combines lexical TF-IDF features with semantic MPNet sentence embeddings, each classified with balanced logistic regression.

TRI AI Saturdays Cohort 10 - Team Gashaka